

Iodine Status and Thyroid Function of Boston-Area Vegetarians and Vegans

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Context: Adequate dietary iodine is required for normal thyroid function. The iodine status and thyroid function of U.S. vegetarians and vegans have not been previously studied. Environmental perchlorate and thiocyanate (inhibitors of thyroid iodine uptake) exposures may adversely affect thyroid function. Objective: The objective of the study was to assess the iodine status and thyroid function of U.S. vegetarians (consume plant based products, eggs, milk; abstain from meat, poultry, fish, shellfish) and vegans (avoid all animal products) and whether these may be affected by environmental perchlorate and thiocyanate exposures. Design and Setting: This was a cross-sectional assessment of urinary iodine, perchlorate, and thiocyanate concentrations and serum thyroid function in Boston-area vegetarians and vegans. Subjects: One hundred forty-one subjects (78 vegetarians, 63 vegans) were recruited; one vegan was excluded. Results: Median urinary iodine concentration of vegans (78.5 µg/liter; range 6.8–964.7 µg/liter) was lower than vegetarians (147.0 µg/liter; range 9.3–778.6 µg/liter) ($P < 0.01$). Adjusted for cigarette smoking (confirmed by urinary cotinine levels) and thiocyanate-rich food consumption, median urinary thiocyanate concentration of vegans (630 µg/liter; range 108–3085 µg/liter) was higher than vegetarians (341 µg/liter; range 31–1963 µg/liter) ($P < 0.01$). There were no between-group differences in urinary perchlorate concentrations ($P = 0.75$), TSH ($P = 0.46$), and free T4 ($P = 0.77$). Urinary iodine, perchlorate, and thiocyanate levels were not associated with TSH ($P = 0.59$) or free T4 ($P = 0.14$), even when adjusted for multiple variables. Conclusions: U.S. vegetarians are iodine sufficient. U.S. vegans may be at risk for low iodine intake, and vegan women of child-bearing age should supplement with 150 µg iodine daily. Environmental perchlorate and thiocyanate exposures are not associated with thyroid dysfunction in these groups.